Junto Mabuchi

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Education

Dulwich College Shanghai — Shanghai, China
International Baccalaureate, Achieved Score of 44/45 — *Aug. 2018 – July 2022*

Imperial College London — London, UK
Bachelor's in Electronic and Information Engineering — *Sep. 2022 – Present*

- Achieved 2:1 Grade in 1st Year

Experience

IC Hack — February 2025
Imperial College London — *London, UK*

- Participated in the largest student-run Hackathon in Europe.
- Developed Cloud9, an AI-powered mental well-being companion leveraging Anthropic AI.
- Built a Next.js frontend and a Express.js backend with PostgreSQL
- Implemented Spotify API integration for music mood matching based on user emotions
- Deployed the project on Vercel (frontend) and Render (backend)

Work Experience at Espressif Systems — Sep. 2018 – Oct. 2019
espressif.com — *Shanghai, China*

- Developed a full-stack web application to track company servers
- Utilized the Django framework and SQLite for the backend
- Collaborated with 4 other students and presented this to the IT department at Espressif

Projects

C Compiler | *C, Git* — January 2025 – March 2025

- Worked with a partner to design a C compiler as part of coursework, translating C code into RISC-V assembly.
- Developed a lexer, parser, and code generator to support key language features, including control flow, functions, and memory management.
- Utilized Git and GitHub for version control

RISC-V CPU | *SystemVerilog, Git* — Nov. 2024 – Dec. 2024

- Designed and implemented a RV32I pipelined RISC-V processor in SystemVerilog as part of coursework
- Integrated a hazard detection unit and cache to support efficient pipeline execution.
- Developed branch prediction & register renaming systems to improve performance
- Collaborated with 3 teammates using Git and GitHub for version control and effective project coordination.

Technical Skills

Languages: Python, C/C++, HTML/CSS, JavaScript SystemVerilog
Frameworks: Django, FastAPI
Developer Tools: Git, VS Code, PyCharm
Libraries: pandas, NumPy, Matplotlib